

MARGARET LAKE, YT, Canada

WMO: 719770

Lat: **68.801N**Lon: **140.847W**Elev: **568**StdP: **94.68**Time Zone: **-8.00 (NAP)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.7	-38.7	-44.7	0.0	-40.4	-42.7	0.1	-38.8	6.7	-19.1	5.5	-20.3	0.3	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	22.9	14.5	21.0	13.4	19.3	12.7	15.9	20.9	14.6	19.4	13.6	18.1	3.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	10.5	17.6	12.5	9.7	16.4	11.3	8.9	15.6	46.6	20.9	42.8	19.6	39.8	18.2	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.5	3.9	DB	-42.2	26.0	3.1	1.5	-44.4	27.1	-46.3	28.0	-48.0	28.8	-50.3	29.9	(4)
(5)			WB	-42.5	16.7	3.0	1.1	-44.6	17.5	-46.4	18.1	-48.0	18.7	-50.2	19.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.9	-26.1	-25.0	-20.9	-11.6	2.0	10.4	12.1	8.5	1.1	-10.9	-21.7	-25.4	(6)	
	DBStd	15.51	7.39	6.87	6.52	6.25	6.54	4.37	3.33	3.87	4.66	5.75	6.30	6.19	(7)	
	HDD10.0	7063	1119	979	957	648	256	46	19	76	266	648	952	1097	(8)	
	HDD18.3	9926	1378	1213	1215	898	508	238	193	305	516	907	1202	1355	(9)	
	CDD10.0	181	0	0	0	0	7	59	85	29	0	0	0	0	(10)	
	CDD18.3	1	0	0	0	0	0	1	1	0	0	0	0	0	(11)	
	CDH23.3	36	0	0	0	0	0	17	13	6	0	0	0	0	(12)	
	CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0	(13)	
(14) Wind	WSAvg	1.2	0.8	0.6	0.7	1.1	1.7	2.1	1.9	1.7	1.4	1.0	0.7	0.6	(14)	
(15) Precipitation	PrecAvg	249	12	9	9	7	13	30	46	46	28	22	14	12	(15)	
	PrecMax	313	33	22	23	21	32	62	89	92	55	38	29	29	(16)	
	PrecMin	190	3	2	1	1	3	8	19	12	14	9	2	3	(17)	
	PrecStd	32	7	4	5	4	6	14	13	16	9	8	7	6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.9	-6.3	-1.7	6.3	21.6	25.7	24.8	23.9	14.8	3.5	-5.9	-8.3	(19)	
		MCWB	-6.6	-7.3	-3.4	2.2	11.7	16.2	15.7	16.2	8.4	2.2	-6.8	-8.8	(20)	
	2%	DB	-10.9	-9.4	-6.4	3.5	18.6	23.1	22.6	20.6	12.5	0.3	-8.6	-12.3	(21)	
		MCWB	-11.3	-10.0	-7.6	0.5	10.6	14.8	14.3	13.7	7.7	-0.5	-9.2	-12.8	(22)	
	5%	DB	-13.1	-12.7	-9.0	1.1	15.3	20.9	20.8	18.1	10.3	-1.3	-10.6	-14.5	(23)	
		MCWB	-13.4	-13.1	-9.9	-1.0	9.2	13.5	13.6	11.9	6.6	-2.1	-11.1	-15.0	(24)	
	10%	DB	-15.9	-15.5	-11.7	-1.1	12.3	19.0	19.3	15.9	8.2	-2.9	-12.4	-16.7	(25)	
		MCWB	-16.2	-15.8	-12.5	-3.0	7.6	12.4	13.2	11.0	5.7	-3.8	-12.9	-17.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.5	-6.9	-3.0	2.7	12.7	18.2	17.1	17.0	9.6	2.3	-6.8	-8.8	(27)	
		MCDB	-5.6	-6.4	-1.6	5.5	19.4	24.0	22.7	23.4	13.2	3.2	-5.7	-8.4	(28)	
	2%	WB	-11.4	-9.7	-7.3	0.8	11.2	16.1	15.8	14.4	8.1	-0.4	-9.1	-12.8	(29)	
		MCDB	-11.2	-9.3	-6.2	3.1	17.6	20.7	20.4	19.6	11.1	0.1	-8.5	-12.5	(30)	
	5%	WB	-13.6	-12.8	-9.9	-0.9	9.8	14.4	14.8	12.9	6.9	-2.1	-11.0	-14.9	(31)	
		MCDB	-13.3	-12.5	-9.0	1.3	15.4	19.6	19.3	17.3	10.0	-1.3	-10.6	-14.5	(32)	
	10%	WB	-16.6	-15.7	-12.5	-2.9	7.9	13.1	13.9	11.4	5.8	-3.9	-12.8	-16.9	(33)	
		MCDB	-16.4	-15.3	-11.8	-1.0	12.4	17.9	18.2	15.3	8.1	-3.1	-12.4	-16.6	(34)	
(35) Mean Daily Temperature Range	MDBR		7.5	8.0	10.3	12.3	11.5	11.8	10.6	10.4	8.6	7.4	6.9	7.4	(35)	
	5% DB	MCDBR	10.4	10.1	11.4	13.3	15.5	14.6	13.8	14.5	12.6	7.8	8.4	9.9	(36)	
		MCWBR	10.3	9.6	9.2	10.4	8.9	7.2	6.9	7.6	7.9	6.8	7.8	9.8	(37)	
	5% WB	MCDBR	10.2	10.0	10.4	12.8	14.9	13.6	12.2	12.6	11.3	7.9	8.1	10.3	(38)	
		MCWBR	10.0	9.6	9.2	10.2	9.2	7.5	6.7	6.9	7.3	6.9	7.9	10.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.160	0.202	0.240	0.296	0.325	0.335	0.361	0.323	0.280	0.211	0.148	N/A	(40)	
	taud		1.902	2.147	2.246	2.149	2.237	2.417	2.379	2.494	2.490	2.185	1.918	N/A	(41)	
	Ebn at Noon		81	622	813	843	859	863	821	822	772	583	78	0	(42)	
	Edh at Noon		5	40	70	107	111	96	97	77	58	36	4	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.03	0.48	2.17	4.55	5.97	6.14	4.97	3.46	1.96	0.67	0.08	0.00	(44)	
	RadStd		0.00	0.04	0.11	0.19	0.39	0.62	0.36	0.32	0.21	0.08	0.01	0.00	(45)	

Historical Trends